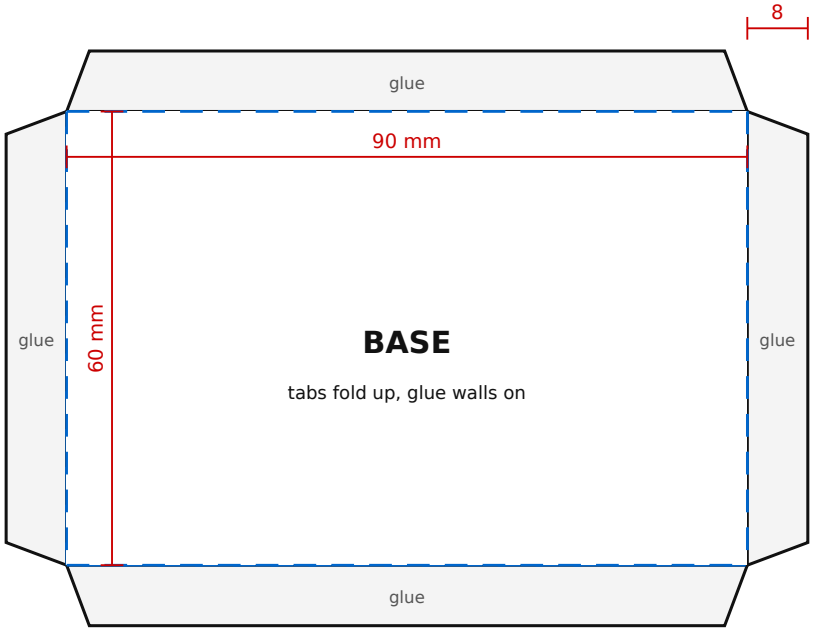
Robotics Project #13 · Templates 1/5 (1:1 scale)

Wheelbarrow Walker Robot

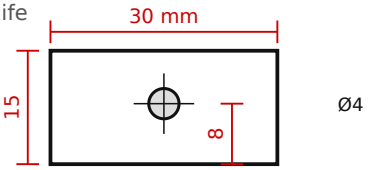
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — — — fold (dashed) ◯ hole: grown-up pokes with a pencil/awl red = size in mm

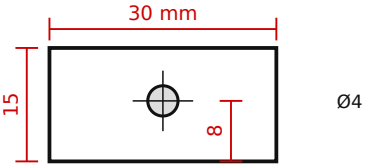
3 mm corrugated cardboard · grown-up cuts with craft knife



Barrow base 90x60 + 8 mm glue tabs (wheels roll freely)

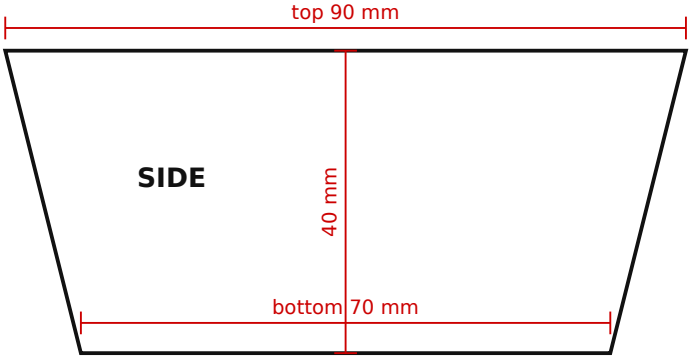


Axle hanger 1

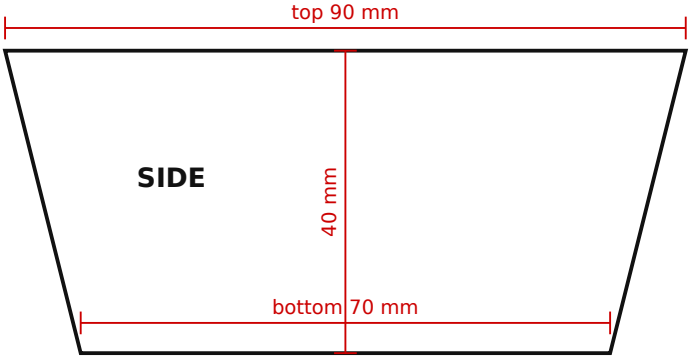


Axle hanger 2

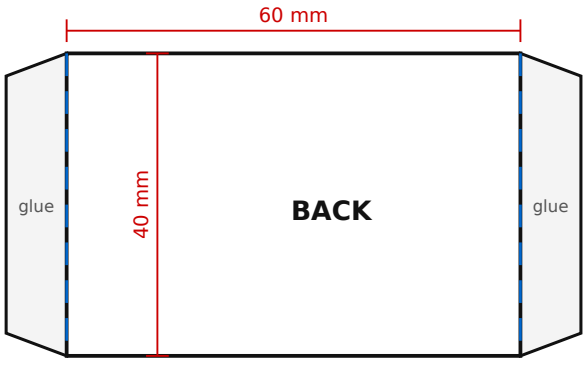
hole 8 mm up from bottom edge



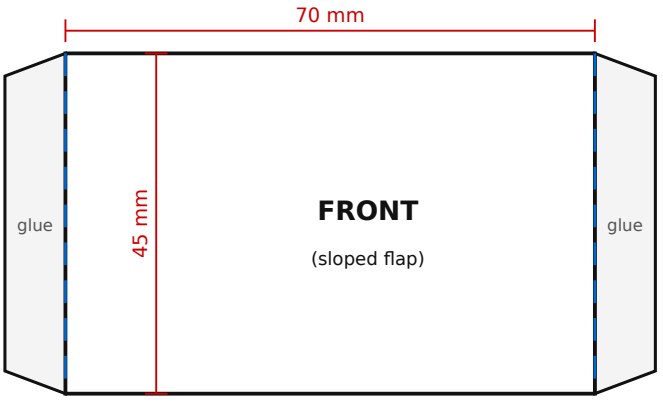
Side wall 1 (trapezoid)



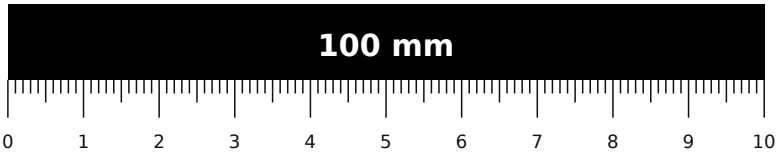
Side wall 2 (trapezoid)



Back wall 60x40 + tabs



Front flap 70x45 + tabs



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

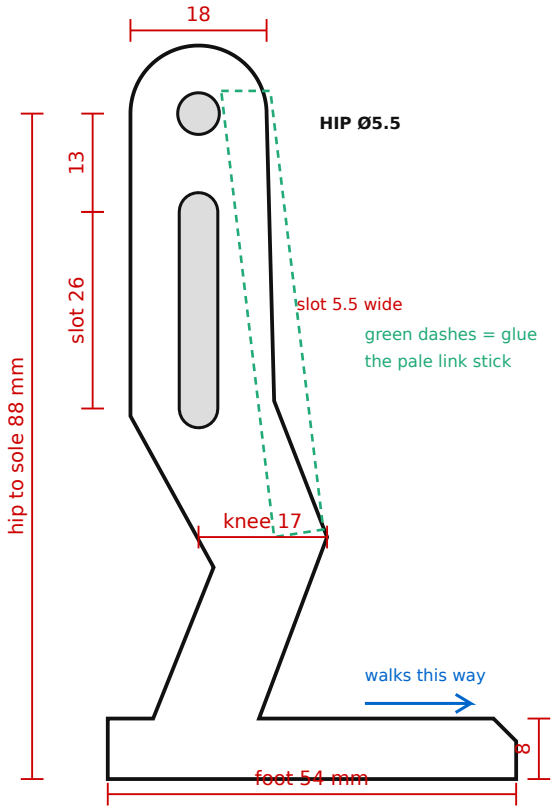
Robotics Project #13 · Templates 2/5 (1:1 scale)

Wheelbarrow Walker Robot

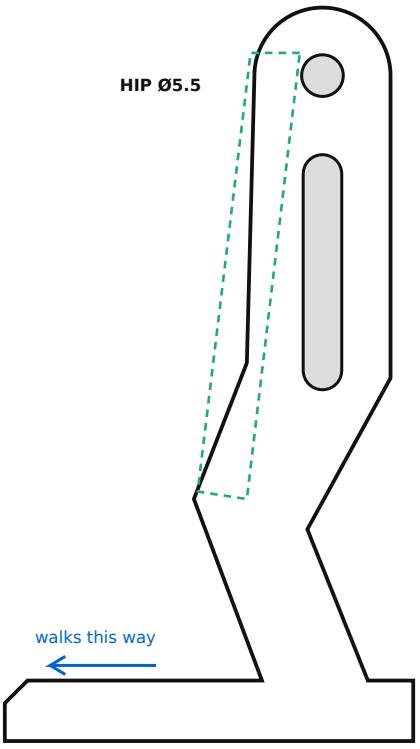
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

3 mm corrugated cardboard. Trace and cut each leg TWICE, glue the two layers = one strong 6 mm leg.



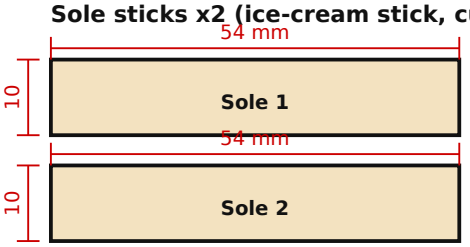
LEFT leg (cut 2)



RIGHT leg (cut 2, mirror)

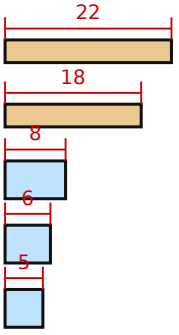
Hip hole, slot and sole must stay on one straight line (the centre line) - that is what makes the foot glide flat.

Sole sticks x2 (ice-cream stick, cut)



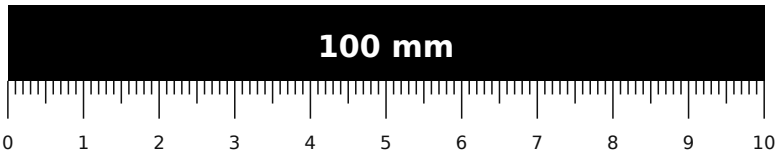
Glue a sole stick under each long foot (toe forward).
GRIP: 2 rubber bands round each foot -
the swinging foot pushes, so it must not slide.

Skewer Ø3 + straw Ø5 - true length:



- **Hip peg (skewer) x2**
- **Crank pin (skewer) x2**
- **Hip sleeve (straw) x2**
- **Roller (straw) x2**
- **Spacer (straw) x2**

Wheel axle: skewer 110 mm (as before). Stoppers: 4 cardboard buttons Ø10 (sheet 3) glued on pin/peg ends.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.

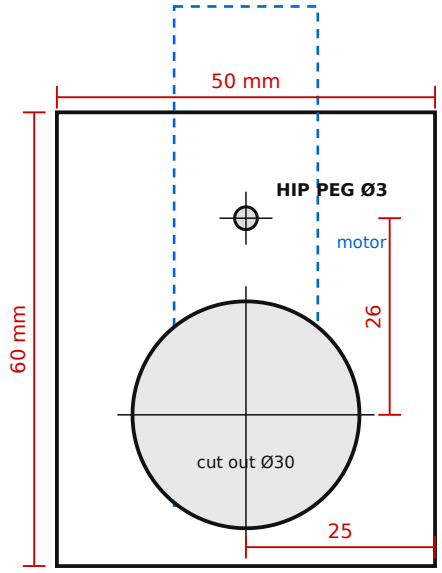
Robotics Project #13 · Templates 3/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

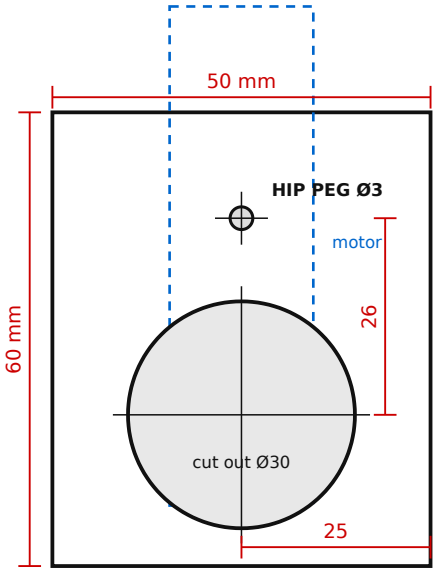
cut (solid) — — — fold (dashed) ○ hole: grown-up pokes with a pencil/awl red = size in mm

3 mm cardboard. Hip plates glue on the two shaft faces of the TT motor (gearbox end DOWN).



Hip plate LEFT

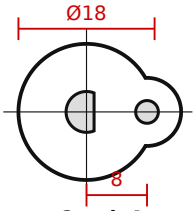
Ø30 round the shaft; blue = motor (66 long)



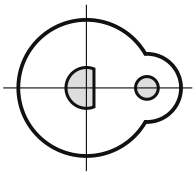
Hip plate RIGHT

Ø30 round the shaft; blue = motor (66 long)

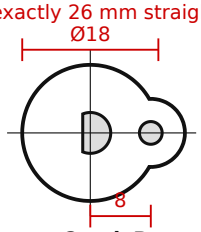
Hip peg exactly 26 mm straight above the shaft - this sets the swing.



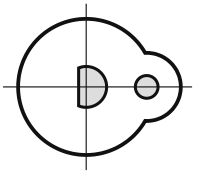
Crank A



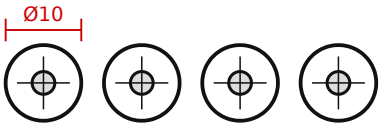
Crank A



Crank B



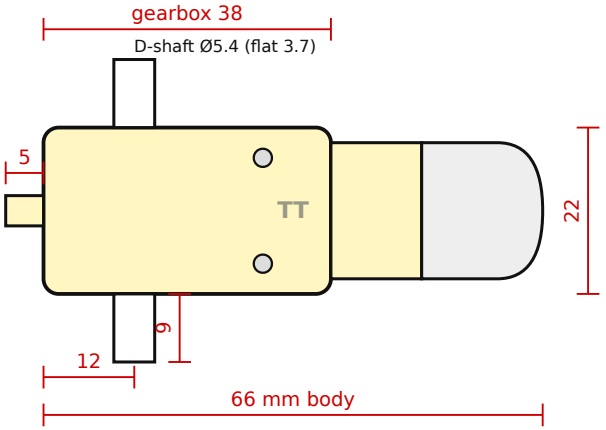
Crank B



Button stoppers x4 (Ø3 hole)

Glue A+A and B+B = two 6 mm cranks. D-hole Ø5.4, 3.7 across the flat: snug on the shaft + hot glue.

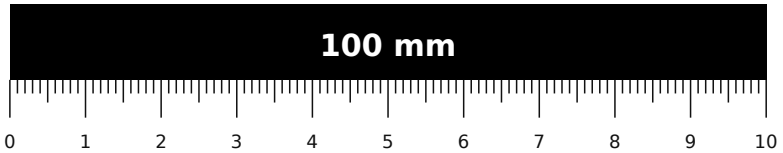
A has the flat on the lobe side, B away from it: same shaft = pins OPPOSITE = 180° apart.



CHECK YOUR MOTOR

Lay it on the drawing: body, both white shafts and the rear nub should match (± 1 mm).

Different? Move the hip peg so it is still 26 mm above the shaft.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

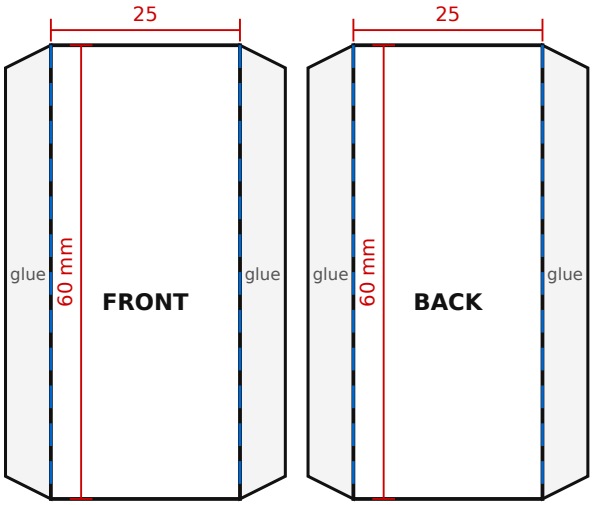
Robotics Project #13 · Templates 4/5 (1:1 scale)

Wheelbarrow Walker Robot

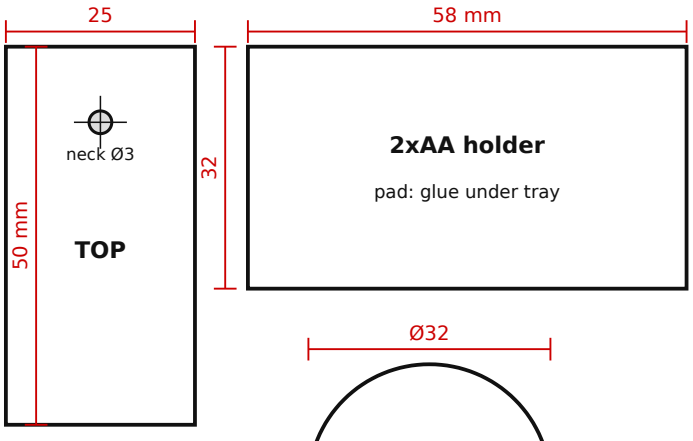
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

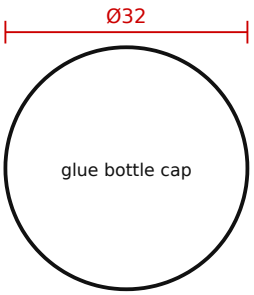
cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Torso covers, battery pad, hat - and a 1:1 side view to check your leg before gluing.

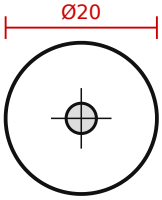


Torso covers 25x60 (+6 mm tabs glue to plate edges)

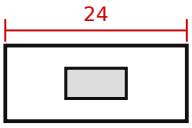




Hat brim



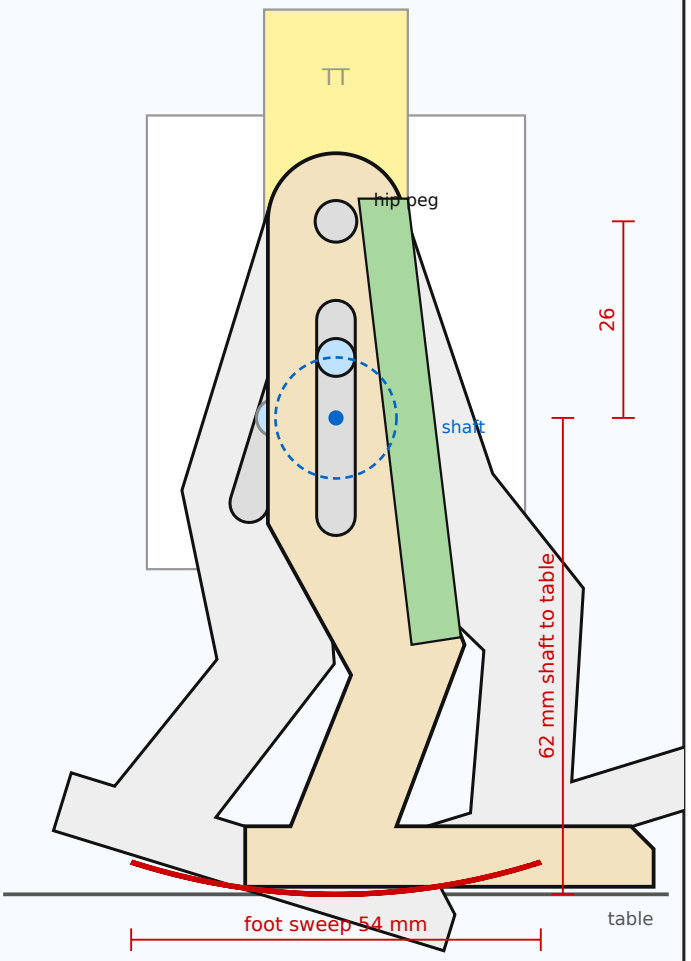
Wheel washer x2 (Ø4)



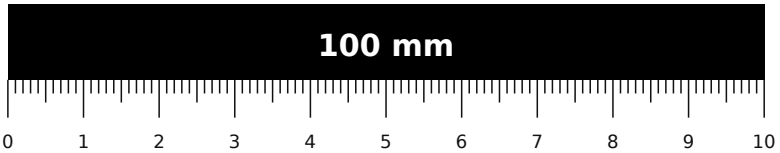
Switch plate (cut 8x4 slot)

1:1 SWING CHECK (side view)

lay your leg on this drawing



Pale legs = crank pin at front / back.
Red = foot path: 54 mm swing, only 4 mm rise.
Walks backwards? Swap the 2 motor wires.



20 mm

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If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.

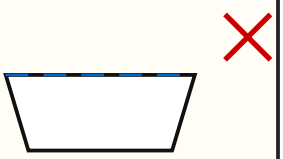
Robotics Project #13 · Templates 5/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

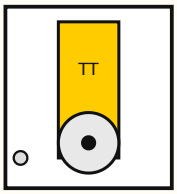
1 Cut and fold the barrow tray (sheet 1). Glue the tabs inside.



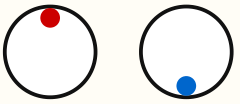
2 Skewer axle through a straw under the tray; glue a wheel on each end. Wheels roll freely.



3 Hot-glue a hip plate (sheet 3) on each flat side of the TT motor, motor upright, shaft in the Ø30 hole.



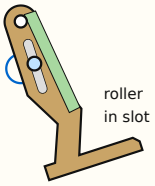
4 Glue crank A on one shaft end, crank B on the other: the pins must point opposite ways.



5 Glue a 22 mm hip peg in each hip hole. Add a 5 mm straw spacer, then the leg (hip sleeve in its Ø5.5 hole).



6 Crank pin + straw roller sits in the leg slot. Glue button stoppers on peg and pin ends.



7 Glue torso covers, arms at the shoulders, hands onto the handles with both feet on the table.

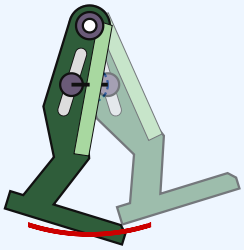


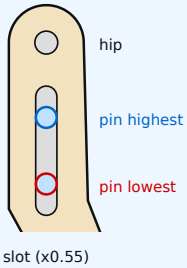
8 2xAA + switch under the tray, wires along a handle to the motor. Switch on and he walks!



ON = walk!

How the legs walk





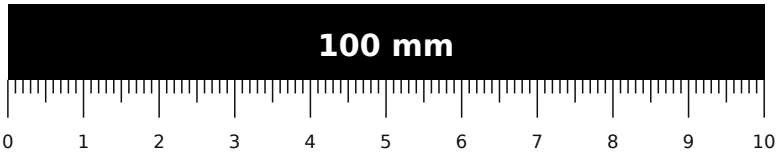
The motor turns both cranks, half a turn apart. The crank pin (with a straw roller) slides up and down in the leg slot and pushes the leg front and back. The leg swings like a pendulum from the hip peg: the foot (red) sweeps 54 mm along the table and rises only 4 mm. One foot pushes back while the other swings forward - and his hands push the barrow! Roller never leaves the slot.

Parts checklist

- ☐ 3 mm cardboard
- ☐ 3 ice-cream sticks
- ☐ 2 wheels (caps/65 mm)
- ☐ Skewers 3 mm
- ☐ Paper straw
- ☐ TT motor (double shaft)
- ☐ 2xAA holder + batteries
- ☐ Slide switch + wire
- ☐ Rubber bands (feet)
- ☐ Ping-pong ball + cap
- ☐ Hot glue (grown-up)
- ☐ Pencil, scissors

Safety first!

- A grown-up uses the hot glue gun, craft knife, and pokes the holes.
- Snip the sharp skewer tips off.
- Fingers away from the cranks when ON.
- Batteries the right way round; never join the wires; take them out after.
- Keep small parts from little kids.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.

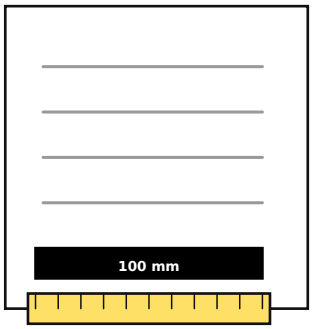
Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

1 Print and check



PARTS All 5 template sheets

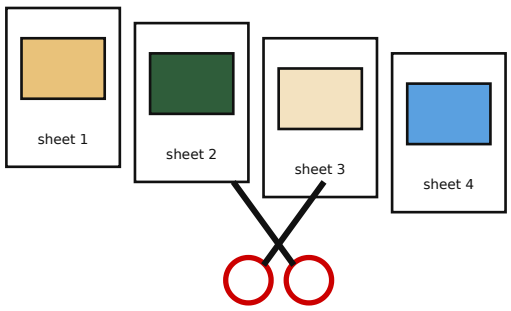
TOOLS Printer, ruler

DO **Print at 100% (Actual size), never 'fit to page'.**

CHECK ✓ The black bar on each sheet measures exactly 100 mm.

2 Cut out the parts

Grown-up !



PARTS Sheet 1 tray, sheet 2 legs, sheet 3 plates & cranks, sheet 4 covers

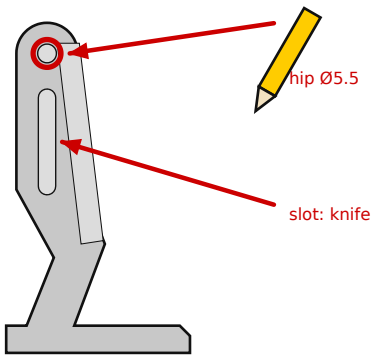
TOOLS Scissors, craft knife

DO **Glue each sheet on cardboard, then cut on the solid lines.**

CHECK ✓ You have 4 leg layers, 2 plates, 4 cranks, 4 buttons.

3 Poke the holes

Grown-up !

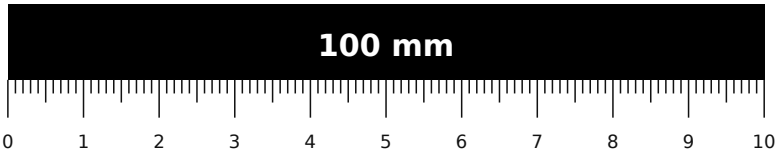


PARTS Sheet 2: Leg L x2, Leg R x2

TOOLS Pencil/awl, craft knife

DO **Poke the Ø5.5 hip holes; knife out the 5.5 mm slots.**

CHECK ✓ A straw slides into the hip hole; a straw fits the slot.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

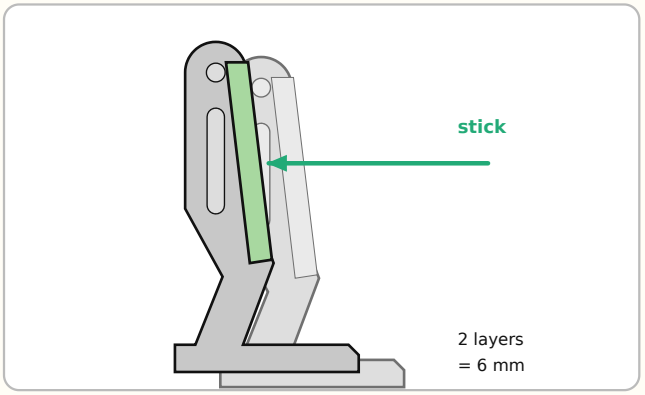
Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

4 Glue the legs



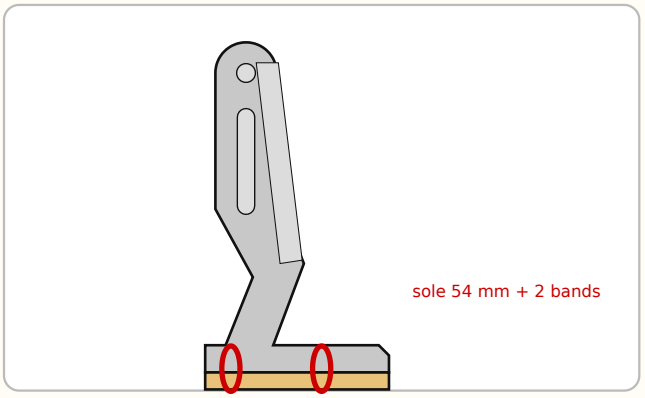
PARTS Sheet 2: Leg layers x4 + 2 ice-cream sticks

TOOLS Glue

DO **Glue 2 layers per leg, then the pale stick along the thigh front.**

CHECK ✓ Holes and slots line up; leg is flat and stiff.

5 Add the long feet



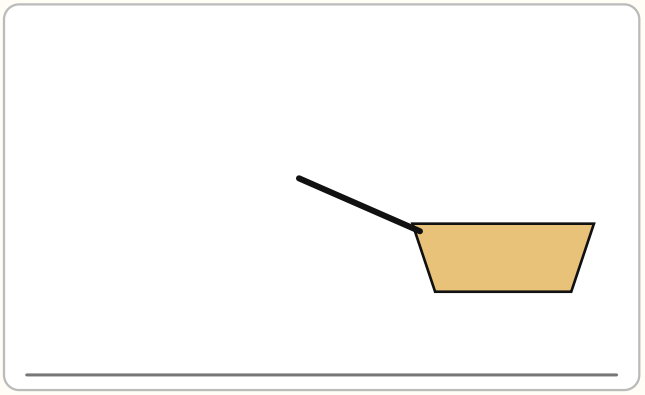
PARTS Sheet 2: Sole sticks x2 (54x10)

TOOLS Glue, rubber bands

DO **Glue a sole stick under each foot, toe forward; add 2 bands.**

CHECK ✓ Both feet lie flat; toe points the walking way.

6 Fold the barrow tray

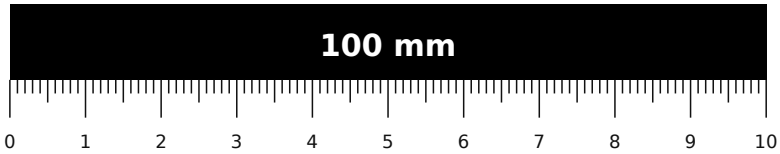


PARTS Sheet 1: tray + side walls

TOOLS Ruler, glue

DO **Score the dashed lines, fold up and glue tabs inside.**

CHECK ✓ Tray box stands square and does not wobble.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

Wheelbarrow Walker Robot

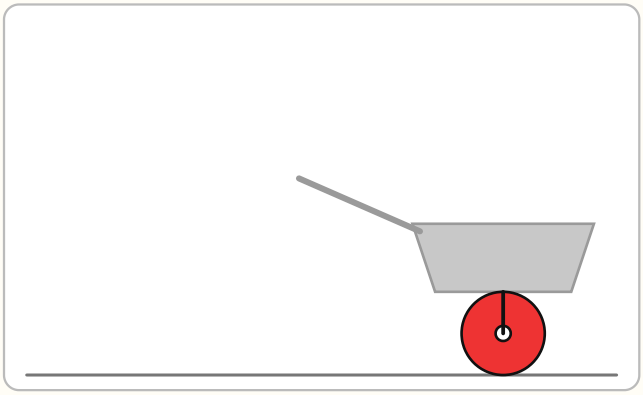
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

7 Axle and wheels

Grown-up !



PARTS Sheet 1 hangers, skewer 110 mm, straw, 2 wheels

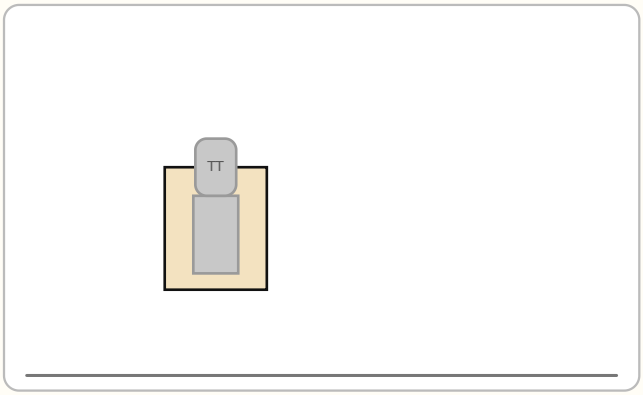
TOOLS Hot glue

DO **Straw under tray, skewer through, wheel glued on each end.**

CHECK ✓ Spin the wheels: they roll freely.

8 Hip plates on motor

Grown-up !



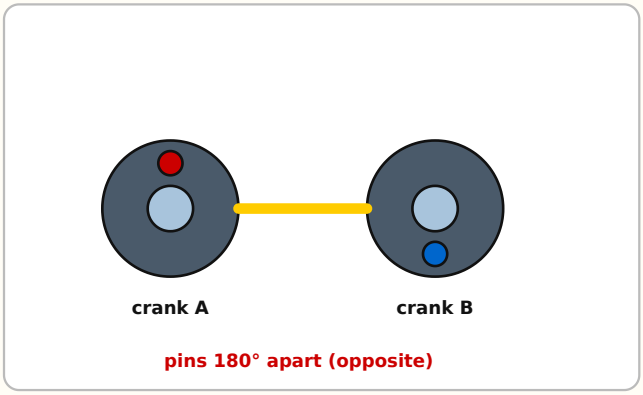
PARTS Sheet 3: hip plates x2, TT motor

TOOLS Hot glue

DO **Motor upright (can on top); plate on each side, shaft in Ø30 hole.**

CHECK ✓ Shaft is in the hole centre; hip hole is straight above.

9 Cranks A then B

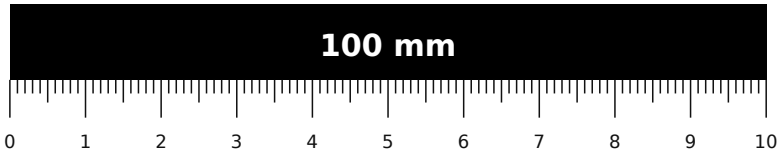


PARTS Sheet 3: crank A x2, crank B x2

TOOLS Glue

DO **Glue crank A on one shaft end, then B on the other side.**

CHECK ✓ Look from the side: pins point opposite (180°).



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

Wheelbarrow Walker Robot

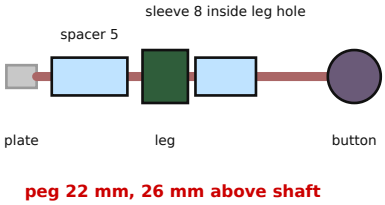
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

10 Hip pegs + spacers

Grown-up !



peg 22 mm, 26 mm above shaft

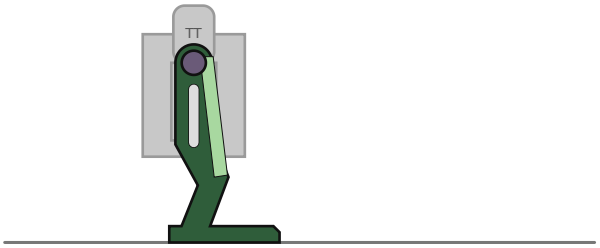
PARTS Skewer 22 mm x2, straw 5 mm + 8 mm

TOOLS Hot glue

DO **Glue a peg in each hip hole, 26 mm above shaft; slip on the spacer.**

CHECK ✓ Peg sticks out straight, not tilted.

11 Hang the left leg



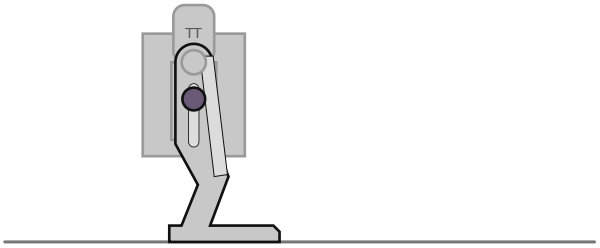
PARTS Leg L, hip sleeve, button

TOOLS Glue

DO **Slide the leg (sleeve in hip hole) on the left peg; glue a button on the outside.**

CHECK ✓ Leg swings freely, button stops it falling off.

12 Pin and roller

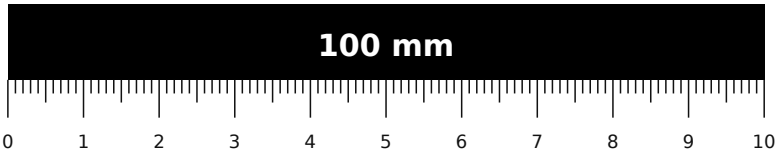


PARTS Skewer 18 mm, straw roller 6 mm, button

TOOLS Glue

DO **Glue the crank pin, slip the roller on, put it in the slot, button outside.**

CHECK ✓ Roller slides along the slot without catching.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

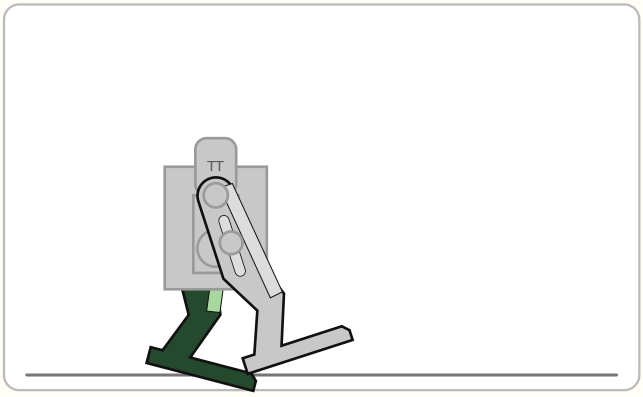
Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

13 Right leg the same



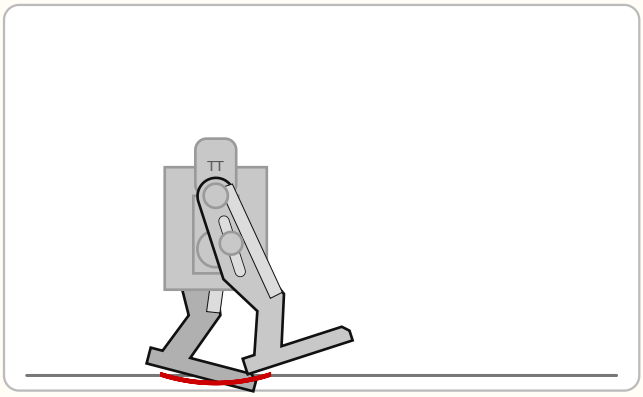
PARTS Leg R, sleeve, roller, 2 buttons

TOOLS Glue

DO **Repeat steps 11-12 on the other side (crank B).**

CHECK ✓ Legs hang opposite: one front, one back.

14 Hand-turn test



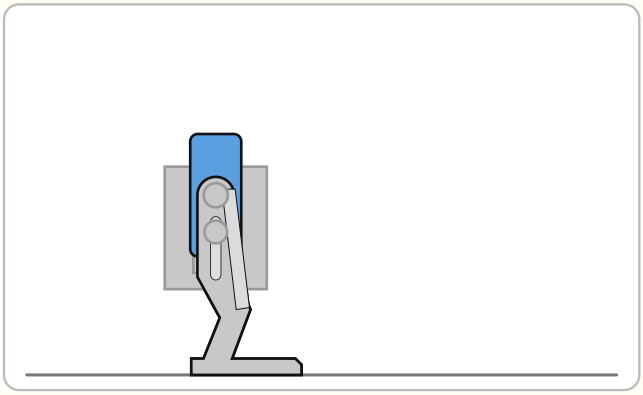
PARTS Sheet 4: 1:1 swing check

TOOLS Fingers

DO **Turn the shaft slowly by hand for 3 full turns.**

CHECK ✓ No jams; foot sweeps about 54 mm like sheet 4.

15 Torso covers

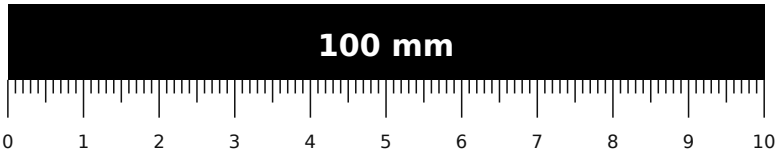


PARTS Sheet 4: FRONT + BACK covers

TOOLS Glue

DO **Glue the covers on the plate edges with the side tabs.**

CHECK ✓ Cranks and legs still turn freely.



20 mm

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If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

Wheelbarrow Walker Robot

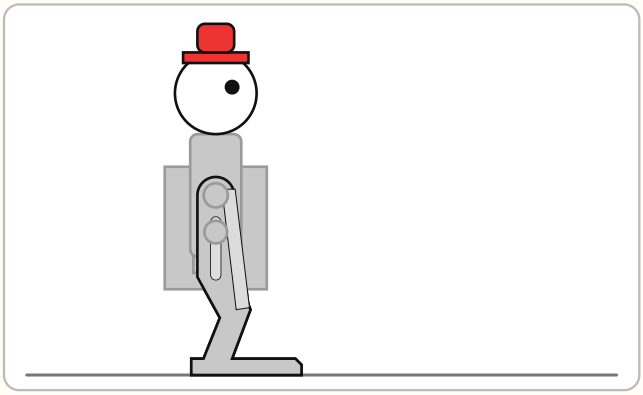
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

16 Head and hat

Grown-up !



PARTS Ping-pong ball, sheet 4 hat brim + bottle cap

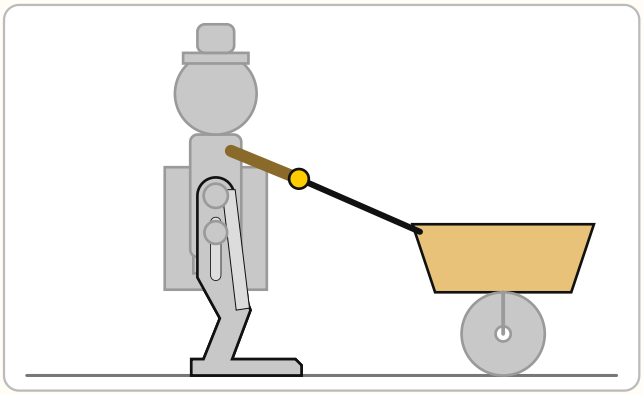
TOOLS Hot glue

DO **Glue the ball on the neck, brim and cap on top as a hat.**

CHECK ✓ Head sits straight; draw a face.

17 Arms to handles

Grown-up !



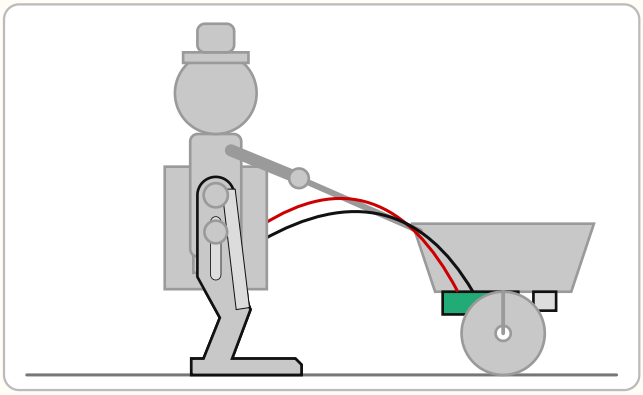
PARTS Ice-cream stick arms, barrow

TOOLS Hot glue

DO **Stand both feet on the table, glue hands onto the handles.**

CHECK ✓ Barrow wheels touch the table; feet flat.

18 Battery and switch

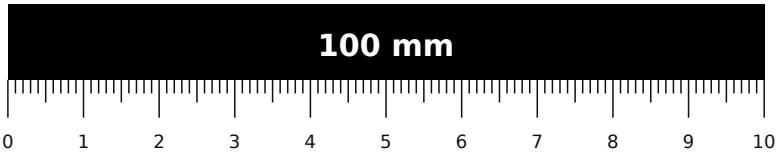


PARTS 2xAA holder, switch plate (sheet 4), wires

TOOLS Tape, glue

DO **Holder + switch under tray; wires along a handle to the motor.**

CHECK ✓ Wires do not touch the legs or cranks.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

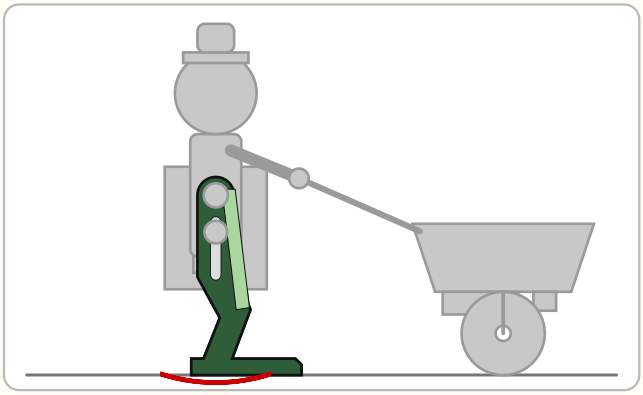
Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Build one step at a time. Colour = the new part, grey = already done. ! = grown-up step.

19 First walk!



PARTS Whole robot

TOOLS Table

DO **Switch ON. Walks backwards? Swap the 2 motor wires.**

CHECK ✓ He pushes the barrow forward, wheels roll.

20 Troubleshooting

Leg jams?
slot not on centre line - re-cut

Feet slip?
more rubber bands

Walks back?
swap the 2 wires

Wobbles?
glue plates square

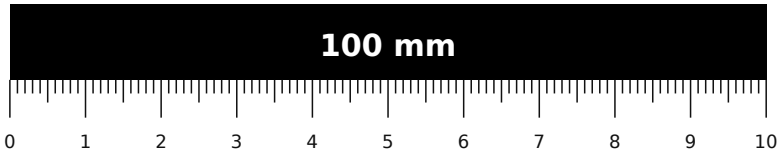
No buzz?
check batteries/switch

PARTS -

TOOLS -

DO **Fix the most common problems:**

CHECK ✓ Still stuck? Re-check sheet 4 swing.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.